CHEM 101 Report for Laboratory Exercise #1

Instructions:

a) Download the .docx file (do not download the pdf file from Brightspace). Using Microsoft Word or equivalent software, insert responses in all yellow highlighted areas.

b) It is recommended the report be completed without changing font size, column width, row width, margins, and highlights. Keep the yellow highlighted areas.

c) Additional instructions are provided throughout the report in the gray highlighted areas. Please read and follow these instructions.

d) All answers must be the student's own work without assistance from others.

e) The completed report must be uploaded to the CHEM 101 Brightspace site as a .pdf file by the due date posted on Brightspace. The report is due to 48 hours of the end of the laboratory period. Late reports will NOT be accepted.

f) ***Only typed reports completed using the template will be graded****. Handwritten reports (or excerpts) will NOT be accepted or graded.*

	Grade Scheme	**max**
Pre-lab quiz	ALL questions answers are correct	2
Laboratory Notebook:	ALL data, procedures and observations accurately recorded	2

Report	Abstract is succinct, and contains objective of the experiment, methods used and some conclusions.	1
	Data, Results followed the outlined in the in-lab notes. Graph, Tables, Calculations, Reported significant figures,	3
	Algebraic equations are correct.	1
	Discussion questions are ALL completed and answered accurately.	2
	Conclusions (a) summarize the most important findings of the exercise; (b) techniques used; (c) interpretation of their significance; (d) addresses the objective.	1
	References: section is complete and correctly formatted, and in-text citation is correct.	1
	Participation: Understand the procedure and engage with the content.	0.5
	Performance: Student followed the safe practices guidelines without exception. Drawers checked. Submission confirmation shown.	0.5
	Maximum Report Grade	10

© 2025, Department of Chemistry, Faculty of Science

CHEM101 Report for Laboratory Exercise #1

Measurement of Volumes and Weights: Accuracy and Precision[1]

Additional Instructions: The number **1** after the text above indicates that more information about the content can be found in **Reference #1 (check if it is complete!)**, which is listed in the reference section at the bottom of the report. Adding a superscript number after the text that refers to a reference is called an **in-text citation**. In your future reports, add superscript numbers corresponding to the references at appropriate points where the referenced information is discussed throughout your report.

Name: Evan O'Toole **Lab Section:** B07 **Quad:** 3 **Date:** September 10th 2025

Abstract

The masses of fixed volumes of distilled water were measured using different glassware. Considering the density of water at the temperature of the experiment, the %error of each piece of glassware was determined. The accuracy and precision of each glassware was evaluated; Results indicated that the volumetric pipette has better accuracy among the tested glassware. Additionally, a graph of density versus concentration of NaCl was plotted, and the concentration of a solution of 1.06 ± 0.02 g/cm^3 was determined to be 0.497 mol/L.

Data/Results

Table 1. Masses, average mass, standard deviation and %RSD using a 100.0 mL graduated cylinder, a 10.00 mL volumetric pipette and a 50.0 mL Beaker for measuring 10 mL of distilled water at room temperature: 22.9°C

	Mass (g) of 10 mL of distilled water					
Glassware	**Sample # 1 (Evan O'Toole)**	**Sample # 2 (Vidhi Parekh)**	**Sample # 3 (Stephanie Park)**	**Average mass (g)**	**Standard Deviation (g)**	**% RSD**
Graduated cylinder	9.680g	9.945g	9.237g	9.621g	0.3577g	3.718%
Volumetric pipette	9.043g	9.970g	9.953g	9.655g	0.5304g	5.494%
Beaker	8.450g	9.961g	9.961g	9.457g	0.8724g	9.225%

Table 2. Nominal volume (volume labeled) by each piece of glassware used for measuring the mass of water, calculated volume and %error. Density of water at 22.9 °C: 0.9975485 g/cm^3

Glassware	**Nominal volume measured (mL)**	**Average volume delivered (mL)**	**%Error**
Graduated cylinder	10.0mL	9.597mL	4.03%
Volumetric pipette	10.00mL	9.631mL	3.690%
Beaker	10mL	9.434mL	5.7%

Table 3. Sodium chloride solutions, masses, densities and error (instrument used: 10.00 mL volumetric pipette) at room temperature: 23.1°C .

	Mass of 10.00 mL solution	**Density (g/cm^3)**
Solution A (0.500 mol ·L^{-1})	10.010g	1.001 g/cm^3
Solution B (0.200 mol ·L^{-1})	10.660g	1.066 g/cm^3
Solution C (0.100 mol ·L^{-1})	10.144g	1.014 g/cm^3

Graph

Additional Instructions: Please see Chapter IV of the lab manual on how to format graphs.

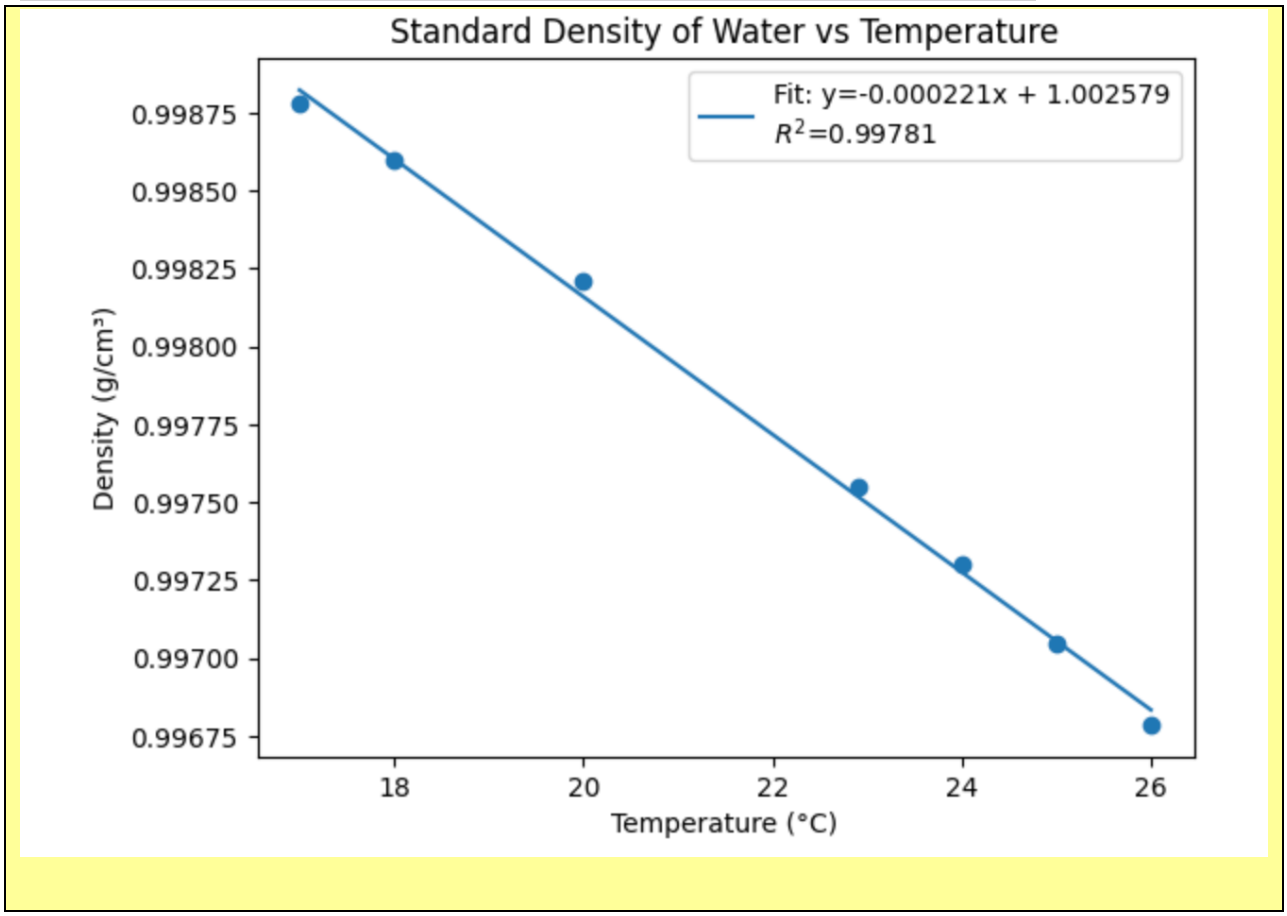

Figure 1. Standard density of water as a function of the temperature.

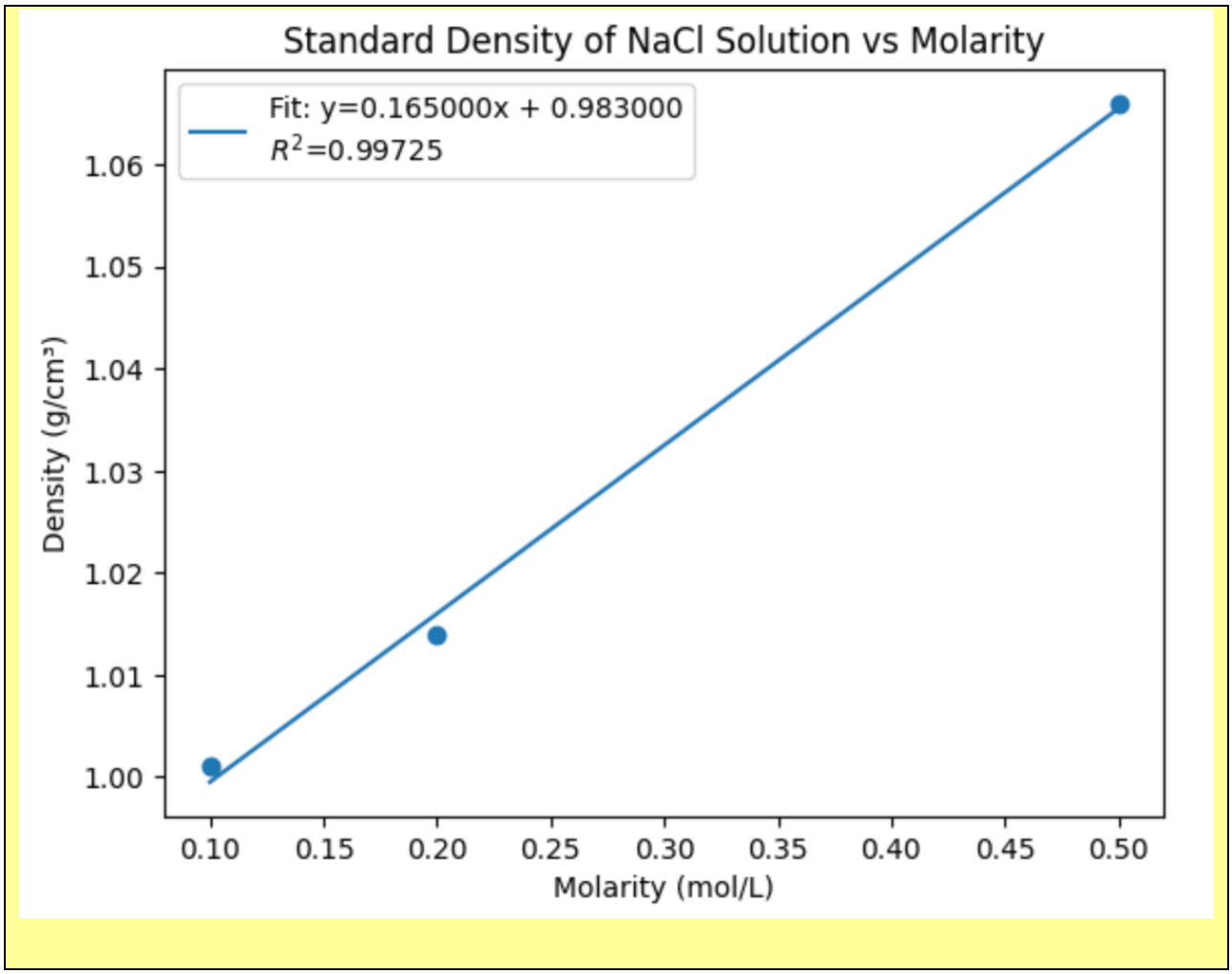

Figure 2. Density of sodium chloride solutions as a function of the molarity. Room temperature 23.1°C

Algebraic Equations

Additional Instructions: Chapter IV of the lab manual.

1) Density

P(g/cm^3) = m(g)/V(cm^3)

Where:

P = Density

m = Mass

V = Volume

2) Mass of NaCl required to prepare a defined molar concentration by dissolution.

$m_{NaCl}(g) = M_{NaCl}(mol/L) \times MW_{NaCl}(g/mol) \times V_{sol}(L)$

Where:
$m_{NaCl}(g)$ = mass of NaCl required
$M_{NaCl}(mol/L)$ = molar concentration
$V_{sol}(L)$ = volume of solvent
$MW_{NaCl}(g/mol)$ = molar weight of NaCl

3) Volume of NaCl initial solution required to prepare a defined final molar concentration by dilution.

$V_{i,NaCl}(L) = (M_{f,NaCl}(mol/L) \times V_{f,sol}(L))/ M_{i,NaCl}(mol/L)$

Where:
$V_{i,NaCl}(L)$ = the volume of initial NaCl solution,
$M_{f,NaCl}(mol/L)$ = final desired molarity of diluted solution,
$M_{i,NaCl}(mol/L)$ = initial molarity of NaCl solution,
$V_{f,sol}(L)$ = final volume of diluted solution

Discussion

1) Based on the results reported in Table 2, discuss which glassware has better accuracy and better precision.

Based on Table 2, the volumetric pipette had the best accuracy and precision. Its accuracy can be observed by comparing the average volume delivered from each glassware to the true value of 10.0mL. The volumetric pipette had the smallest difference from the true value, followed by the graduated cylinder, and then the beaker. This ranking indicates that the volumetric pipette is most accurate. Similarly, precision can be compared using the standard deviation of each measurement. The volumetric pipette had the smallest standard deviation, followed by the graduated cylinder, and lastly the beaker. This indicates that the volumetric pipette was most precise, followed by the graduated cylinder, and finally the beaker.

2) How does the addition of sodium chloride affect the density of water?
In Figure 2 and Table 3, we observe that the density of water linearly increases as the amount of sodium chloride we add to the solution increases.

3) Considering that the density of NaCl aqueous solution as a function of concentration is approximately linear at low to moderate concentrations, and employing the results obtained in the linear regression in Figure 2, estimate the concentration of a NaCl solution which the density measured was 1.065 g·cm^{-3}. Explain your reasoning.
We are given our y variable in the form y = mx + b. we know the values of m, y and, b, and we want to solve for the value of x (mol/L). so rearrange the equation: (y-b)/m = x. plugging in our values gives us that x = 0.4970 mol/L.

4) Discuss the intermolecular forces present in pure distilled water and in an aqueous NaCl solution. Compare and evaluate the relative strengths of these forces
Pure distilled water is composed of polar molecules that form hydrogen bonds between the hydrogen of one water molecule and the oxygen of another. These hydrogen bonds are moderately strong intermolecular forces but aren't as strong as the ones between ions and water molecules present when NaCl is added to water. When NaCl dissolves in water, it dissociates into its component Na+ and Cl- ions. The polar water molecules surround these ions, forming strong intermolecular attraction between the ions and water molecules.

5) What are the key procedural and conceptual differences when preparing a solution by dissolution versus by dilution? (up to 5 lines maximum)
Conceptually, dissolution involves creating a solution using raw solute, where dilution reduces the concentration of an existing solution. Procedurally, dissolution requires weighing the solute and dissolving it in a solvent to the desired volume. Alternatively, dilution requires measuring part of a concentrated solution and adding solvent until it reaches a desired concentration.

Conclusions *(up to 5 lines maximum)*
Additional Instructions: Please see Chapter IV of the lab manual on how to write conclusions.
In conclusion, the analysis of our experimental data indicated that the volumetric pipette was most accurate and precise, followed by the graduated cylinder, and finally the beaker. Additionally, the data suggests an approximately linear increase between density and NaCl concentration. Limitations of this experiment include the inexperience of the experimenters, and a small dataset.

References

1. Lemos M., Reimer M., et al, *Chemistry 101 – Fundamentals of Chemistry from atoms to materials Laboratory Manual*, pp. 16-31. (University of Victoria: Victoria, B.C.). **Fall 2025**.